

Feedback Delay Networks For Reverb Effects

Warren L. G. Koontz

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1 Introduction

Reverberation, or simply reverb, is a popular effect to apply to an audio recording. It provides the impression that the audio has been recorded in an acoustically live, but hopefully not too live, space. In such a space, sound travels from the source to the listener over multiple paths either directly or through one or more reflections. At a given moment, the listener is hearing “newer” sounds that follow a direct path from the source and “older” sounds that follow indirect paths. The oldest sounds, which have lost power but are still audible, may have originated several seconds before the newest sounds.

A reverb effect algorithm processes a sequence of audio input samples to create a corresponding sequence of audio output samples. If each input sample only affects a single output sample, there will be no reverb effect. But if input sample $x(n)$ can affect output sample $y(n)$ as well as a few seconds of future output samples, reverb is a possibility. The feedback delay network (FDN) is one way to do this.

The FDN is described in the following sections based on two components: a delay line with frequency-dependent loss and a standard state space filter. This version of an FDN has been coded as a C++ class called FDNreverb. This document is provided along with the code to provide some theoretical background for users.

2 Delay Line With Loss

Audio signal processing algorithms often include sample delays ranging from a few samples for a simple filter to thousands of samples to simulate an echo or reverberation. In the latter cases, the sample delay is a consequence of the time required for sound to propagate along a path between the source to the listener. For a path of

length L , the time required is simply

$$T = L/c \quad (1)$$

where c is the speed of sound. If the sampling rate is f_s Hz (samples/second), then the corresponding sample delay is

$$M = \text{round}(f_s T) \quad (2)$$

The basic equation for a sample delay is simply

$$y(n) = x(n - M) \quad (3)$$

where $x(n)$ and $y(n)$ are sequences of input and output samples, respectively and M is the sample delay. If we apply the z-transform, we obtain an equally simple expression:

$$Y(z) = z^{-M} X(z) \quad (4)$$

Equation (3) and (4) model a basic delay line.

A frequency-dependant loss can be modeled as a simple low-pass filter given by

$$y(n) = dy(n - 1) + gx(n) \quad (5)$$

or, in terms of a transfer function

$$\begin{aligned} H(z) &= Y(z)/X(z) \\ &= \frac{g}{1 - dz^{-1}} \end{aligned} \quad (6)$$

The frequency response of the low-pass filter is given by

$$H(\Omega) = \frac{g}{1 - de^{-j\Omega}} \quad (7)$$

where Ω is the frequency in radians/sample. For $\Omega = 0$ (corresponding to frequency $f = 0$ Hz), the magnitude of the frequency response is

$$|H(0)| = \frac{g}{1 - d} \quad (8)$$

For $\Omega = \pi$ (corresponding to frequency $f = f_s/2$) the magnitude of the frequency response is

$$|H(\pi)| = \frac{g}{1 + d} \quad (9)$$

If the loss is due to absorption of sound by the medium, then the loss at high frequencies is considerably larger than at low frequencies. Thus we will assume that $|H(0)| = 1$, i.e., negligible loss at low frequency. Under this assumption, $g = 1 - d$ and

$$|H(\pi)| = \frac{1 - d}{1 + d} \quad (10)$$

Under these assumptions, we will refer to the parameter d as the *damping*. The high-frequency attenuation $|H(\pi)|$ might also be expressed in terms of a high-frequency absorption A_{HF} expressed in dB/meter. Then the attenuation in dB for a path of length L is simply $A_{\text{HF}}L$ and the attenuation is given by

$$|H(\pi)| = 10^{-A_{\text{HF}}L/20} \quad (11)$$

From Equations (10) and (11) we obtain the following equation for the damping as a function of the high frequency absorption and the path length:

$$d = \frac{1 - 10^{-A_{\text{HF}}L/20}}{1 + 10^{-A_{\text{HF}}L/20}} \quad (12)$$

Under all of these assumptions, the frequency response of the low-pass filter is

$$H(\Omega) = \frac{1 - d}{1 + de^{-j\Omega}} \quad (13)$$

with the damping d given by Equation (12). Figure 1 is a plot of the frequency response for $L = 20$ meters and $A_{\text{HF}} = 0.3$ dB/meter. Figure 2 is a plot of the impulse response.

Putting the low-pass filter together with the simple delay, we get the following model for a delay line with frequency-dependant loss:

$$y(n) = dy(n - 1) + (1 - d)x(n - M) \quad (14)$$

or

$$H(z) = \frac{(1 - d)z^{-M}}{1 - dz^{-1}} \quad (15)$$

Note that the parameters d and M are both determined from the high-frequency absorption A_{HF} , the path length L , the speed of sound c , and the sampling frequency f_s . The frequency response of the delay line is the same as that of the low-pass filter and the impulse response differs only by a time delay.

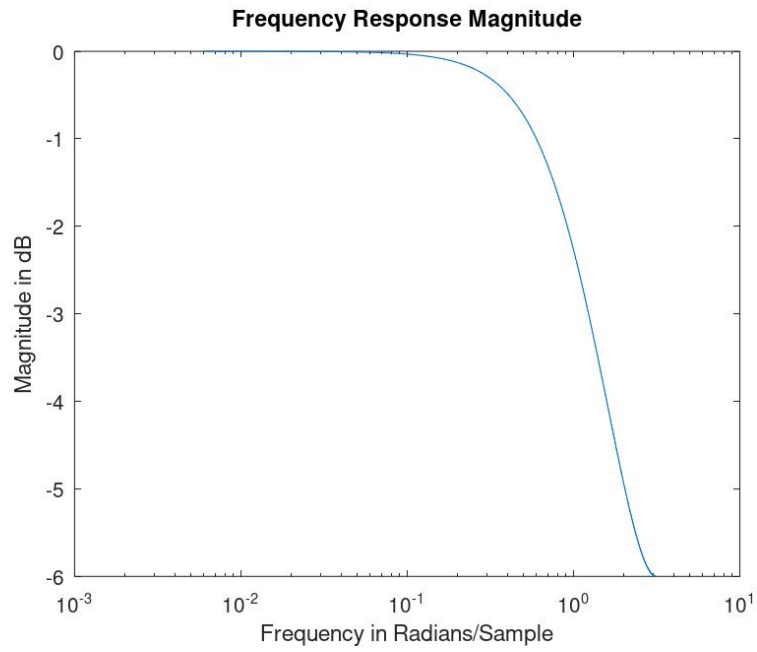


Figure 1: Frequency Response of Low-Pass Filter

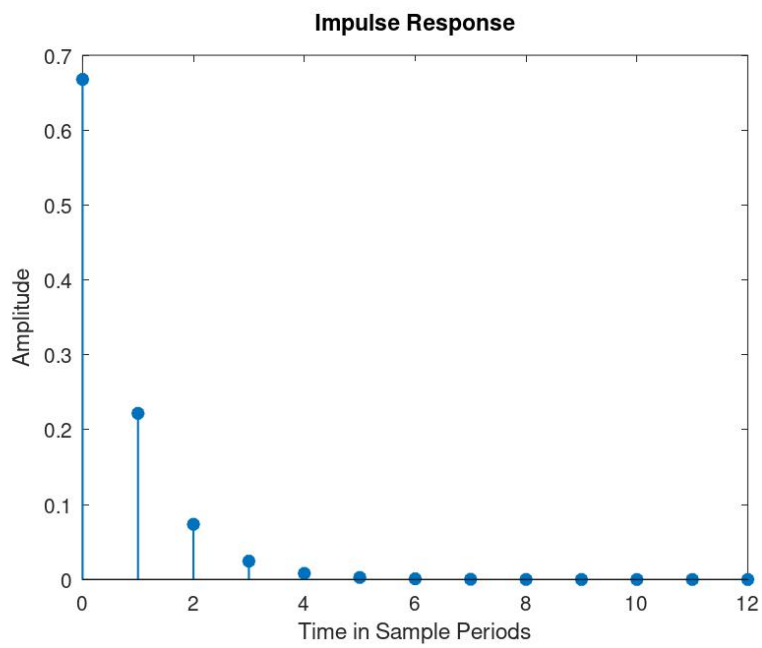


Figure 2: Impulse Response of Low-Pass Filter

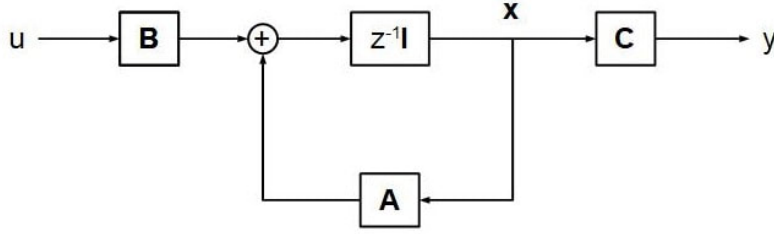


Figure 3: Basic State Space Model

3 Feedback Delay Network

The FDN can be described as an extension of the standard state space model of a linear time-invariant system (LTIS). For discrete time systems with a single input stream and a single output stream, the state space model is

$$\begin{aligned}\mathbf{x}(n+1) &= \mathbf{A}\mathbf{x}(n) + \mathbf{B}u(n) \\ y(n) &= \mathbf{C}\mathbf{x}(n) + Du(n)\end{aligned}\tag{16}$$

In this notation $u(n)$ and $y(n)$ are the input and output streams, respectively and the column vector $\mathbf{x}(n)$ is the *state vector*. If the dimension of $\mathbf{x}$ is N , then $\mathbf{A}$ is a N by N state transition matrix, $\mathbf{B}$ is a N by 1 distribution vector, and $\mathbf{C}$ is a 1 by N accumulation vector. In the z domain, Equation (16) becomes

$$\begin{aligned}z\mathbf{x}(z) &= \mathbf{A}\mathbf{x}(z) + \mathbf{B}u(z) \\ y(z) &= \mathbf{C}\mathbf{x}(z) + Du(z)\end{aligned}\tag{17}$$

Figure 3 illustrates the state space model as a feedback system (assuming $D = 0$).

Figure 4 shows an expanded version of the state space model where the one sample delay ($z^{-1}\mathbf{I}$) is replaced by a more general matrix $\mathbf{H}(z)$. To obtain a FDN, let the various matrices be defined as follows:

$$\mathbf{A} = \frac{R}{N}\mathbf{J} - \mathbf{I}\tag{18}$$

$$\mathbf{B} = [1 \quad 1 \quad \cdots \quad 1]^T\tag{19}$$

$$\mathbf{C} = \frac{1}{N} [1 \quad 1 \quad \cdots \quad 1]\tag{20}$$

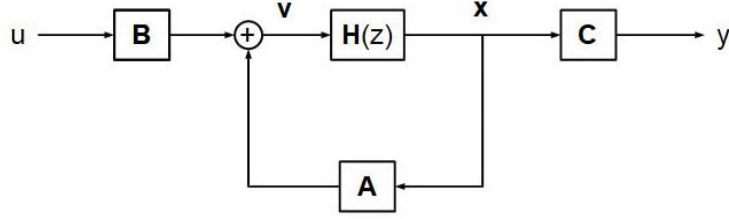


Figure 4: Enhanced State Space Model

where $\mathbf{J}$ is a $N \times N$ matrix of ones and R is a “reflection” parameter. $\mathbf{H}(z)$ is a $N \times N$ diagonal matrix with diagonal elements given by

$$H_{kk}(z) = \frac{(1 - d_k)z^{-M_k}}{1 - d_k z^{-1}} \quad (21)$$

which models a bank of delay lines with loss.

3.1 Frequency Domain Analysis of FDN

The transfer function for the FDN is

$$\begin{aligned} h(z) &= \frac{y(z)}{u(z)} \\ &= \mathbf{C}[\mathbf{I} - \mathbf{H}(z)\mathbf{A}]^{-1}\mathbf{H}(z)\mathbf{B} \end{aligned} \quad (22)$$

The FDN frequency response $h(\Omega)$ is obtained from Equation (22) with $z = e^{-j\Omega}$. Figure 5 shows the magnitude of the frequency response for a reasonable example of a FDN. The complexity of the frequency response reflects the fact that the FDN is a very high-order filter. In fact, the frequency response does not seem particularly useful except for the fact that it can be transformed into the impulse response using the inverse FFT as shown in Figure 6.

3.2 Time Domain Analysis of FDN

For implementation purposes it is more useful to return to the time domain. The relation between the state vector $\mathbf{x}$ and the intermediate vector $\mathbf{v}$ in Figure 4 is

$$x_k(z) = \frac{(1 - d_x)z^{-M_k}}{1 - d_k z^{-1}} v_k(z) \quad (23)$$

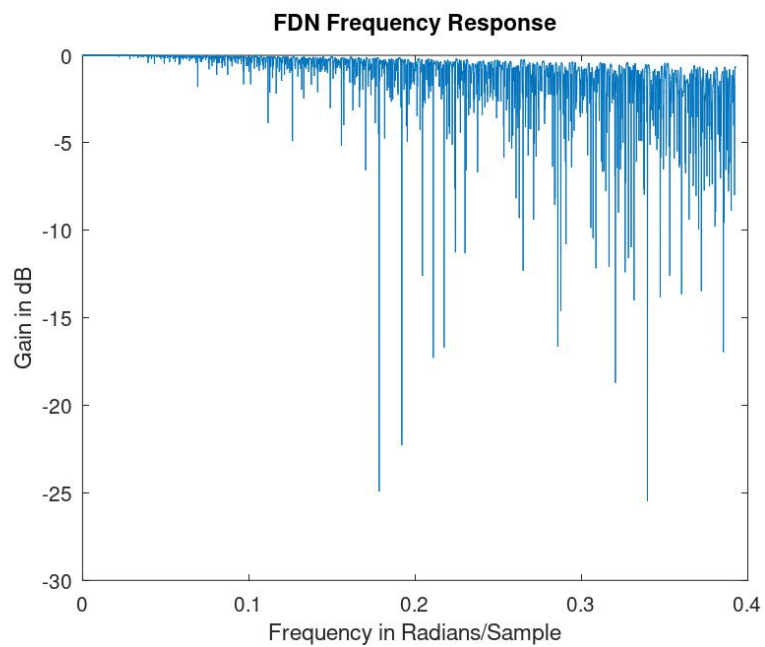


Figure 5: FDN Frequency Responsel

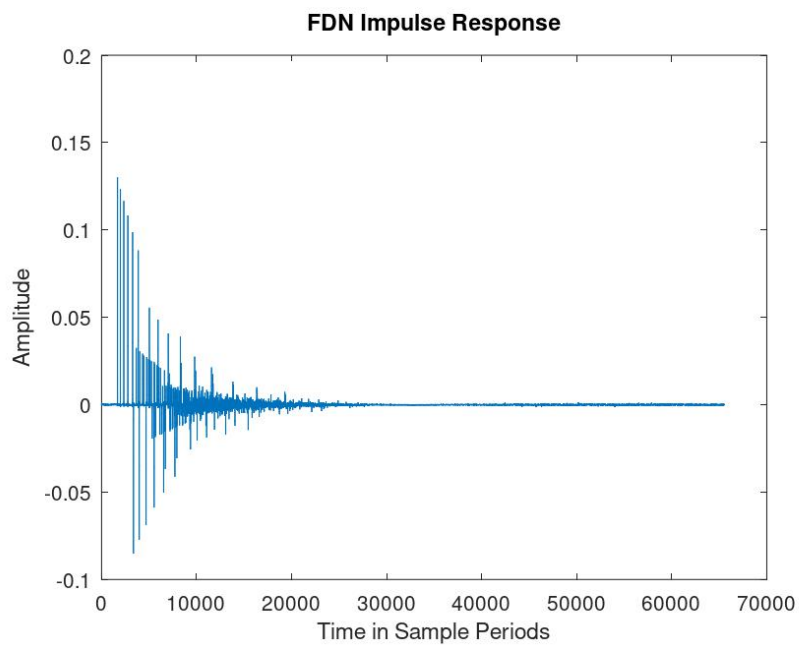


Figure 6: FDN Impulse Responsel

for $k = 0, 1, \dots, N - 1$. In the time domain this becomes a simple difference equation.

$$x_k(n) = d_k x_k(n - 1) + (1 - d_k) v_k(n - M_k) \quad (24)$$

Based on the definitions of **B** and **C**, we have additional relations:

$$y(n) = \frac{1}{N} \sum_{k=0}^{N-1} x_k(n) \quad (25)$$

and

$$\begin{aligned} v_k(n) &= u(n) + \frac{R}{N} \sum_{l=0}^{N-1} x_l(n) - x_k(n) \\ &= u(n) + R y(n) - x_k(n) \end{aligned} \quad (26)$$

Equations (24), (25), and (26) can be used to determine the output sequence $y(n)$ from the input sequence $u(n)$ as follows:

- For $k = 0, 1, \dots, N - 1$, let $x_k(0) = 0$ and $v_k(n) = 0$ for $n < 0$.
- For $n = 0, 1, \dots$
 - Compute $y(n)$ using Equation (25)
 - Compute $v_k(n)$ for $k = 0, 1, \dots, N - 1$ using Equation (26)
 - Compute $x_k(n + 1)$ for $k = 0, 1, \dots, N - 1$ using Equation (24)

This set of equations forms the basis of a FDN reverb algorithm.

4 Implementation Considerations

I have coded an FDN based on Equations (24)–(26) as a C++ class. This section provides an overview of this implementation.

The “order” of the FDN, i.e., the value of N , is set to 6 at compile time. This value seems to be reasonable.

The following parameters are provided to be set by the user:

- Room Area (sq m) - used to determine a maximum path length (m) that is used to determine the sample delays M_k .

- High Frequency Absorption (dB/m) - used to determine the damping values d_k .
- Reflection - the value of R .

The maximum path length, which determines the maximum sample delay, is derived from the room area based on the following, rather flakey reasoning. First assume that the room is a golden rectangle with width W and length ΦW where

$$\Phi = \frac{1 + \sqrt{5}}{2} \quad (27)$$

Since the area of the room is ΦW^2 , the length of the room can be determined from the room area:

$$\begin{aligned} L &= \Phi W \\ &= \Phi \sqrt{A_R / \Phi} \\ &= \sqrt{\Phi A_R} \end{aligned} \quad (28)$$

The sample delays M_k are based on a sequence of path segments ranging in length from L down to $L/10$ exponentially.

$$L_k = \alpha^k L \quad (29)$$

where

$$\alpha^{N-1} = 0.1 \quad (30)$$

The sample delays are given by

$$M_k = \text{round}(f_s L_k / c) \quad (31)$$

where f_s is the sampling rate and c is the speed of sound. The path segment lengths L_k can also be used to compute the damping values.

$$d_k = \frac{1 - 10^{-A_{HF} L_k / 20}}{1 + 10^{-A_{HF} L_k / 20}} \quad (32)$$

where A_{HF} is the high frequency absorption coefficient.

Table 1 provides an example of sample delays and damping values obtained by this procedure. For this example, the room area is 300 sq m, the high frequency absorption is 0.3 dB/m, and the sampling rate is 44100 Hz.

Index	Sample Delay	Damping
0	2833	0.363124
1	1787	0.235558
2	1128	0.150324
3	712	0.095282
4	449	0.060229
5	283	0.038030

Table 1: Example Values of Sample Delays and Damping